

PRESENTATION TALK SHEET

Treasure Island Hurricane Change Detection

Automated pre/post-storm satellite comparison across three independent sources — ORB + RANSAC registration, SSIM structural analysis, and DINOv2 semantic similarity.

HURRICANES

Helene + Milton, 2024

SOURCES

Maxar · NOAA · Sentinel-2

RUN TIME

5 minutes

1 · REGISTER



2 · SSIM



3 · DINOv2



KEY FINDING

Motivation & Study Site

THE PROBLEM

- Manual before/after damage assessment after a hurricane is slow and subjective
- A raw pixel difference gets fooled by things that aren't damage: clouds, tide level, sun angle, exposure
- Goal: an automated pipeline that flags **real** change and stays honest about where it's confused

THE SITE

- Treasure Island, FL — barrier island
- Hit by **Hurricane Helene** (Sept 2024), then **Hurricane Milton** 11 days later (Oct 2024)

THREE INDEPENDENT SOURCES, ON PURPOSE

Maxar

sub-meter · pre-season vs
post-Milton

HIGHEST DETAIL

NOAA NGS

high-res aerial · post-Helene vs
post-Milton

MILTON ONLY

Sentinel-2

10–20 m/px · pre vs post

CLOUD-AFFECTED

Registration — ORB + Lowe's Ratio + RANSAC

Two satellite passes never line up pixel-for-pixel. Everything downstream is meaningless unless the images are aligned first.

HOW IT WORKS, IN ORDER

- 1 ORB finds keypoints** — corner-like points in each image, each turned into a 256-bit binary "fingerprint" (fast to compare)
- 2 Lowe's ratio test filters matches** — for each point, compare its best match vs. second-best; keep it only if the best is clearly better. Kills ambiguous matches on repetitive stuff (waves, rooftops, sand)
- 3 RANSAC fits the final transform** — repeatedly tries small random groups of matches, keeps whichever transform the most other points agree with. Robust even if a few bad matches slipped through step 2
- 4** Output: a **3×3 homography matrix** → warps the "after" image onto the "before" image's frame

CONFIDENCE SCORE — RANSAC INLIER COUNT

140 Maxar inliers	833 NOAA inliers	810 Sentinel-2 inliers
WEAKEST FIT	STRONG	STRONG

SSIM — Structural Similarity

A plain pixel subtraction can't tell "different lighting" from "actual damage." SSIM compares structure, not raw brightness.

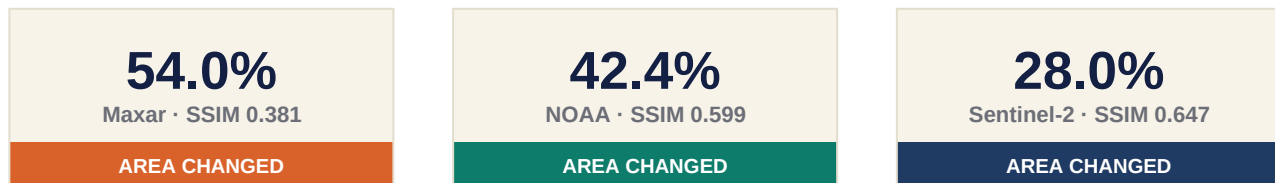
THREE LOCAL COMPARISONS, MULTIPLIED TOGETHER

- **Luminance** — similar average brightness?
- **Contrast** — similar brightness spread?
- **Structure** — similar spatial pattern, after removing brightness/contrast?
- Score bounded $\sim[-1, 1]$, 1.0 = identical
- Invert to $1 - SSIM \rightarrow$ dissimilarity map
- **Otsu's threshold** auto-splits that map into changed/unchanged — no manual tuning
- Reported as **% of frame changed**

WHY IT'S ROBUST

A sunnier day barely moves the structure term (pattern's unchanged) $\rightarrow$ SSIM shrugs it off. A sand berm turning into jagged debris collapses the structure term $\rightarrow$ SSIM flags it.

RESULTS



DINOv2 — Learned Semantic Similarity

SSIM only asks "did the pixels reorganize." It can't tell "sand became water" from "sand looks slightly different." DINOv2 asks a semantic question instead.

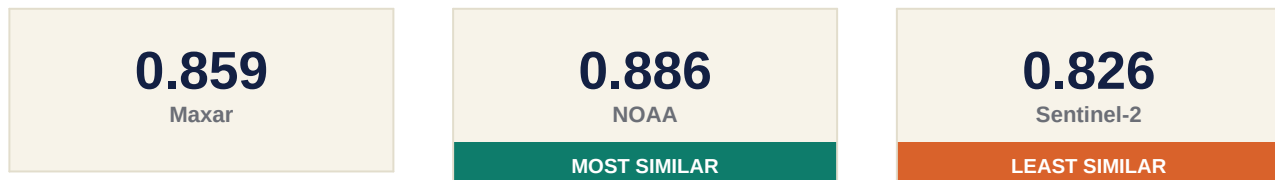
WHAT IT ACTUALLY IS

- Model: **facebook/dinov2-base** — a Vision Transformer
- **Self-supervised, no labels** — trained so different crops/views of the same image produce consistent representations
- Side effect: embeddings organize by *what kind of thing* is in the patch, not pixel stats
- **Pure embedding model** — no classification head, no softmax, no fine-tuning here
- The only math done on top: **cosine similarity** (plain linear algebra, nothing learned)

HOW IT'S USED

- Image → 14×14 patches → one token per patch + one [CLS] token summarizing the whole image
- **Global score**: cosine similarity between the two images' [CLS] tokens
- **Spatial map**: cosine similarity patch-by-patch across a 16×16 grid, upsampled for visualization

RESULTS — GLOBAL COSINE SIMILARITY



The Key Finding — The Disagreement

THE CONSISTENT STORY

All three sources agree: **Gulf-facing beach/dune took the damage, interior canals stayed stable.**

THE INTERESTING EXCEPTION — SENTINEL-2



Same data. Opposite conclusions. Both caused by clouds in the pre-image, not real change.

- SSIM: a cloud is still smooth texture → doesn't disturb structure much → scores it *similar*
- DINOv2: a cloud is a completely different semantic surface than "ground" → scores it *very different*

Neither metric is wrong — they measure different things. Running them together turns a silent false signal into a visible flag: this comparison is confounded, don't trust it at face value.

Limitations & Future Work

KEY LIMITATIONS

- 1 No cross-sensor calibration** — scores are relative, not absolute. 54% (Maxar) is not directly comparable to 28% (Sentinel-2)
- 2 Cloud cover confounds both metrics, in opposite directions** — demonstrated directly by the Sentinel-2 result
- 3 Small sample, no ground truth** — 3 pairs, 1 site, internally consistent but not externally validated

- Maxar's registration (140 inliers) is the weakest of the three — headline result, shakiest foundation
- DINOv2 was pretrained on natural/web images, not satellite data — domain mismatch

FUTURE RESEARCH

- **Validate against ground truth** — compare outputs to real FEMA/NOAA damage surveys, calibrate a real damage scale
- **Swap in a remote-sensing-pretrained model** — SatMAE, Prithvi, or Clay instead of natural-image DINOv2
- **Explicit cloud masking** — use Sentinel-2's cloud-probability bands before differencing
- **Scale up** — more sites, more storms, for statistically meaningful trends
- **Automate the disagreement** — turn SSIM/DINOv2 divergence into a built-in reliability flag
- **Time series, not two frames** — track recovery/rebuilding over months
- **Gate on registration quality** — auto-flag low-inlier-count pairs like Maxar's 140

Combining a classical structural signal with a learned semantic signal, across multiple independent sensors, gives a self-checking pipeline — where they agree, the result can be trusted; where they disagree, that disagreement is itself the finding.